S181M116 Derivatives and Alternative Investments

Lecture 5: Stochastic Calculus

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Today

1. Why continuous-time derivative pricing needs stochastic processes.
2. What the Markov property means for stock prices.
3. How Brownian motion is built from independent normal increments.
4. Why uncertainty scales with the square root of time.
5. How drift and volatility enter a stochastic differential equation.
6. Why stock prices are modeled with geometric Brownian motion.
7. How to simulate stock price paths.
8. How correlated shocks are constructed.
9. Why stochastic calculus has different multiplication rules.
10. How Ito's lemma transforms functions of stochastic variables.
11. Why geometric Brownian motion implies lognormal stock prices.

Core Message

The single most important scale difference

Ordinary calculus studies small changes of order dt. Brownian shocks are of order $\sqrt{dt}$:

$$dz \approx \varepsilon\sqrt{dt}, \qquad \varepsilon \sim \mathcal{N}(0,1).$$

Consequence

Since $(dz)^2$ is of order dt, the square of a random shock does not disappear. This creates the extra $\frac{1}{2}$ term in Ito's lemma.

From Binomial Trees to Brownian Motion

Lecture 4 used a binomial tree:

$$S \to Su \quad \text{or} \quad S \to Sd.$$

Continuous-time finance asks what happens as:

- the number of steps becomes very large;
- each time step becomes very small;
- the up/down random walk converges to a continuous random path.

Bridge

Brownian motion is the continuous-time limit object behind many recombining tree models. It lets us do local hedging and obtain differential equations.

Section 1

Stochastic Processes

1. Stochastic Processes
2. Brownian Motion
3. Drift, Volatility, and Stock Prices
4. Simulation and Correlation
5. Stochastic Calculus
6. Lognormal Stock Prices

What Is a Stochastic Process?

Definition

A stochastic process is a collection of random variables indexed by time:

$$\{X_t : t \geq 0\}.$$

Each X_t is the uncertain value of the variable at date t.

Time

- Discrete time: $t = 0, 1, 2, \ldots$.
- Continuous time: $t \in [0, T]$.

State

- Discrete state: finite/countable values.
- Continuous state: any value in an interval.

Four Types of Processes

	Discrete state	**Continuous state**
Discrete time	Binomial tree	AR(1) return model
Continuous time	Poisson jump process	Brownian stock model

Model versus reality

Real stock prices trade in ticks and only when markets are open. Continuous-time models are approximations chosen because they are tractable and useful.

The Markov Property

Definition

The future depends on the current state, not on the path taken to reach it:

$$\mathbb{P}(X_u \leq x \mid X_t, X_{t_1}, \ldots, X_{t_n}) = \mathbb{P}(X_u \leq x \mid X_t), \qquad t_i < t < u.$$

Stock price interpretation

If the stock price is 100 today, the model's future distribution depends on 100 and current parameters. It does not separately depend on whether the stock was 80 or 120 last month.

▸ Conditional-distribution detail

Markov Does Not Mean History Is Useless

History can help estimate

- volatility;
- jump frequency;
- correlations;
- time-varying parameters.

Markov means

After parameters and current state are known, the past path is not an additional state variable.

Market-efficiency intuition

Predictable patterns in past prices tend to be traded away.

Section 2

Brownian Motion

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Variance and the Square Root of Time

Suppose independent one-year changes satisfy

$$\Delta X_1 \sim \mathcal{N}(0, 1).$$

For two years, the total change is the sum of two independent normal changes:

$$\Delta X_2 = \Delta X_{1,a} + \Delta X_{1,b}.$$

Key rule

Means add and variances add:

$$\Delta X_2 \sim \mathcal{N}(0, 2), \qquad \mathrm{sd}(\Delta X_2) = \sqrt{2}.$$

Derive variance scaling

Volatility Scaling Example

If annual volatility is 20%, then under square-root-time scaling:

$$\sigma_{1\text{ month}} = 20\%\sqrt{\frac{1}{12}} = 5.77\%.$$

$$\sigma_{1\text{ week}} = 20\%\sqrt{\frac{1}{52}} = 2.77\%.$$

Common mistake

Do not divide annual volatility by 12 to get monthly volatility. Divide variance by 12, or divide standard deviation by $\sqrt{12}$.

Wiener Process

Definition

A process z_t is a Wiener process, or standard Brownian motion, if:

1. over a small interval Δt,

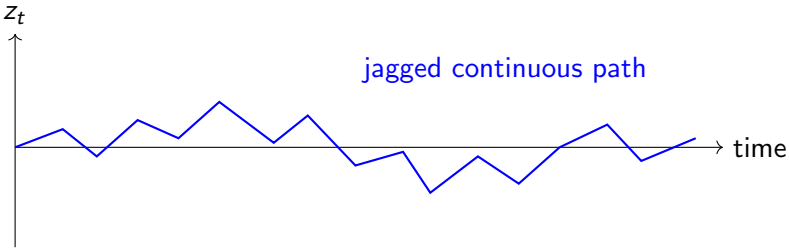
$$\Delta z = \varepsilon\sqrt{\Delta t}, \qquad \varepsilon \sim \mathcal{N}(0,1);$$

2. increments over non-overlapping intervals are independent.

Hence:

$$\mathbb{E}[\Delta z] = 0, \qquad \mathrm{Var}(\Delta z) = \Delta t, \qquad z_T - z_0 \sim \mathcal{N}(0, T).$$

Brownian Motion as a Limit



Path intuition

Brownian motion is continuous, but not smooth. It has no ordinary instantaneous slope because smaller time grids keep revealing new variation.

Why Brownian Paths Are Jagged

Brownian increments satisfy

$$dz \approx \varepsilon\sqrt{dt}.$$

For small dt, $\sqrt{dt}$ is much larger than dt:

$$dt = 0.0001 \quad \Rightarrow \quad \sqrt{dt} = 0.01.$$

Implication

Random movement dominates deterministic drift at very short horizons. This makes Brownian paths locally rough rather than tangent-line smooth.

Section 3

Drift, Volatility, and Stock Prices

1. Stochastic Processes
2. Brownian Motion
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Generalized Wiener Process

A standard Wiener process has zero drift and variance rate one.

Add drift and scale

$$dx = a\,dt + b\,dz.$$

Over a small interval:

$$\Delta x = a\Delta t + b\varepsilon\sqrt{\Delta t}.$$

Therefore:

$$\mathbb{E}[\Delta x] = a\Delta t, \qquad \mathrm{Var}(\Delta x) = b^2\Delta t.$$

Drift and Volatility: Visual Interpretation

Drift a

Moves the center of the distribution:

$$\mathbb{E}[x_T - x_0] = aT.$$

Diffusion coefficient b

Controls dispersion:

$$\mathrm{sd}(x_T - x_0) = b\sqrt{T}.$$

Mental model

Drift moves the center of the cloud. Volatility changes the width of the cloud. A single path can still move against the drift.

Numerical Example: Cash Balance

Cash balance, in thousands, follows

$$dx = 20\,dt + 30\,dz.$$

If $x_0 = 50$, then after one year:

$$x_1 \sim \mathcal{N}(70, 900), \qquad \mathrm{sd}(x_1) = 30.$$

After six months:

$$x_{0.5} \sim \mathcal{N}(60, 450), \qquad \mathrm{sd}(x_{0.5}) = 21.21.$$

Interpretation

The mean moves linearly with time. The standard deviation moves with $\sqrt{T}$.

Ito Process

A generalized Wiener process uses constant a and b.

State-dependent drift and volatility

$$dx = a(x,t)\,dt + b(x,t)\,dz.$$

Over a small interval:

$$\Delta x \approx a(x,t)\Delta t + b(x,t)\varepsilon\sqrt{\Delta t}.$$

Local approximation

In the next instant, evaluate drift and volatility at the current state (x,t). Then update the state and repeat.

Why Stock Prices Need Percentage Shocks

A model like

$$dS = a\,dt + b\,dz$$

gives constant dollar drift and constant dollar volatility.

Problem

A daily standard deviation of 2 dollars is huge for a 10 dollar stock and small for a 200 dollar stock.

Stock returns are naturally proportional:

- expected compensation is quoted as percent return;
- volatility is quoted as percent volatility;
- compounding acts on returns, not dollar changes.

Geometric Brownian Motion

Stock price model

$$dS = \mu S\, dt + \sigma S\, dz, \qquad \frac{dS}{S} = \mu\, dt + \sigma\, dz.$$

μ

Real-world expected continuously compounded return.

σ

Volatility: standard deviation of log returns per square root of year.

▸ Deterministic growth derivation

Discrete-Time Approximation

From

$$\frac{dS}{S} = \mu\, dt + \sigma\, dz,$$

the small-step approximation is

$$\frac{\Delta S}{S} = \mu \Delta t + \sigma \varepsilon \sqrt{\Delta t}.$$

Therefore, approximately:

$$\frac{\Delta S}{S} \sim \mathcal{N}(\mu \Delta t, \sigma^2 \Delta t).$$

Interpretation

Expected return over Δt is $\mu \Delta t$. Return standard deviation over Δt is $\sigma\sqrt{\Delta t}$.

Weekly Return Example

Let

$$\mu = 15\%, \qquad \sigma = 30\%, \qquad \Delta t = \frac{1}{52} = 0.01923.$$

Then

$$\frac{\Delta S}{S} = 0.15(0.01923) + 0.30\varepsilon\sqrt{0.01923} = 0.00288 + 0.0416\varepsilon.$$

Interpretation

Expected weekly return is 0.288%. Weekly return standard deviation is 4.16%.

Section 4

Simulation and Correlation

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Euler Simulation of GBM

The simple simulation update is

$$S_{t+\Delta t} = S_t \left(1 + \mu \Delta t + \sigma \varepsilon \sqrt{\Delta t}\right).$$

1. Choose a time step Δt.
2. Draw $\varepsilon \sim \mathcal{N}(0, 1)$.
3. Update S.
4. Repeat with independent new shocks.

Important

This is an approximation. For GBM there is also an exact lognormal update.

Exact Lognormal Simulation

For geometric Brownian motion with constant μ and σ:

$$S_{t+\Delta t} = S_t \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t}\varepsilon\right].$$

Why use this?

- It keeps prices positive.
- It is exact for GBM.
- It follows from applying Ito's lemma to $\ln S$.

▸ Derive lognormal update

One Simulated Step

Let

$$S_t = 100, \quad \mu = 0.15, \quad \sigma = 0.30, \quad \Delta t = \frac{1}{52}, \quad \varepsilon = 0.52.$$

Euler gives

$$\Delta S = 100(0.00288 + 0.0416(0.52)) = 2.45.$$

The next simulated price is

$$S_{t+\Delta t} = 102.45.$$

Annualizing Volatility

If σ is annual volatility, return standard deviation over T years is

$$\sigma\sqrt{T}.$$

With 252 trading days:

$$\sigma_{\text{daily}} = \frac{\sigma_{\text{annual}}}{\sqrt{252}}, \qquad \sigma_{\text{annual}} = \sigma_{\text{daily}}\sqrt{252}.$$

Example

Annual volatility 32% implies daily volatility

$$\frac{0.32}{\sqrt{252}} = 2.02\%.$$

Correlated Wiener Processes

Suppose

$$dx_1 = a_1 dt + b_1 dz_1, \qquad dx_2 = a_2 dt + b_2 dz_2.$$

If the Brownian shocks have correlation ρ, we write:

$$\mathrm{Corr}(dz_1, dz_2) = \rho, \qquad dz_1 dz_2 = \rho\, dt.$$

Financial uses

Multi-asset options, spread options, stochastic volatility models, interest-rate models, and risk simulations.

Construct correlated normals

Example: Correlated Shocks

Draw independent $u, v \sim \mathcal{N}(0,1)$ and set

$$\varepsilon_1 = u, \qquad \varepsilon_2 = \rho u + \sqrt{1-\rho^2}\, v.$$

If $\rho = 0.60$, $u = 1.10$, and $v = -0.40$, then:

$$\varepsilon_1 = 1.10, \qquad \varepsilon_2 = 0.60(1.10) + 0.80(-0.40) = 0.34.$$

Interpretation

ε_2 has a common component with ε_1 plus independent residual noise.

Section 5

Stochastic Calculus

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The Multiplication Rules

Since

$$dz \approx \varepsilon\sqrt{dt},$$

we use the informal table:

$$(dt)^2 = 0, \qquad dt\,dz = 0, \qquad (dz)^2 = dt.$$

For correlated Brownian motions:

$$dz_i dz_j = \rho_{ij} dt.$$

Key intuition

The square of the Brownian shock survives because it is order dt.

▸ Why $(dz)^2 = dt$?

Ito's Lemma

Let

$$dx = a(x,t)dt + b(x,t)dz, \qquad G = G(x,t).$$

Then

$$dG = \left(G_x a + G_t + \frac{1}{2} G_{xx} b^2 \right) dt + G_x b \, dz.$$

Compared with ordinary calculus

The extra term is

$$\frac{1}{2} G_{xx} b^2 dt.$$

It comes from curvature times variance.

Derive Ito's lemma

Financial Interpretation of Ito's Lemma

Let $G(S,t)$ be an option value and S the underlying stock.

$$dG = \left(G_S \mu S + G_t + \frac{1}{2} G_{SS} \sigma^2 S^2 \right) dt + G_S \sigma S \, dz.$$

Delta

G_S: first-order exposure to the stock.

Theta

G_t: direct effect of time passing.

Gamma

G_{SS}: curvature interacting with variance.

Example: Squaring Brownian Motion

Let $x = z$, so $dx = dz$, and take

$$G(x) = x^2.$$

Since

$$G_x = 2x, \qquad G_{xx} = 2,$$

Ito's lemma gives

$$d(x^2) = 2x\,dz + dt.$$

Ordinary calculus would miss the dt term

The extra dt is accumulated Brownian variance.

▸ Power-function example

Application: Forward Price Dynamics

For a non-dividend-paying stock and constant r, the forward price at time t for maturity T is

$$F(S,t) = Se^{r(T-t)}.$$

If

$$dS = \mu S dt + \sigma S dz,$$

then Ito's lemma gives

$$dF = (\mu - r)F dt + \sigma F dz.$$

Interpretation

The forward price has the same volatility as the stock and real-world drift $\mu - r$.

Derive forward dynamics

Section 6

Lognormal Stock Prices

1. Stochastic Processes
2. Brownian Motion
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Applying Ito to $\ln S$

If

$$\frac{dS}{S} = \mu dt + \sigma dz,$$

then applying Ito's lemma to $G(S) = \ln S$ gives

$$d \ln S = \left(\mu - \frac{1}{2}\sigma^2\right) dt + \sigma dz.$$

Ito correction

Expected log growth is lower than expected arithmetic instantaneous return by

$$\frac{1}{2}\sigma^2.$$

▸ Derive $d \ln S$

Why Volatility Lowers Log Growth

Equal percentage gains and losses do not offset in compounded wealth.

Simple arithmetic

A 50% gain followed by a 50% loss:

$$100 \to 150 \to 75.$$

The arithmetic average return is zero, but wealth falls.

Continuous-time version

The correction $-\frac{1}{2}\sigma^2$ captures this compounding effect in the GBM model.

Terminal Distribution

Integrating

$$d \ln S = \left(\mu - \frac{1}{2}\sigma^2\right) dt + \sigma dz$$

from 0 to T:

$$\ln S_T \sim \mathcal{N}\left(\ln S_0 + \left(\mu - \frac{1}{2}\sigma^2\right) T, \sigma^2 T\right).$$

Conclusion

S_T is lognormally distributed. Log prices are normal; prices are positive and right-skewed.

Distribution derivation

Mean versus Median

Under real-world GBM:

$$\mathbb{E}[S_T] = S_0 e^{\mu T}.$$

The median is

$$\text{median}(S_T) = S_0 e^{(\mu - \frac{1}{2}\sigma^2)T}.$$

Why they differ

The lognormal distribution is right-skewed. Rare large upward outcomes raise the mean above the median.

Numerical Distribution Example

Let

$$S_0 = 100, \qquad \mu = 12\%, \qquad \sigma = 30\%, \qquad T = 1.$$

Then

$$\ln S_1 \sim \mathcal{N}(\ln 100 + 0.075, 0.09).$$

$$\mathbb{E}[S_1] = 100e^{0.12} = 112.75.$$

$$\text{median}(S_1) = 100e^{0.075} = 107.79.$$

Preview: Risk-Neutral Dynamics

Real-world stock dynamics:

$$\frac{dS}{S} = \mu dt + \sigma dz.$$

Risk-neutral stock dynamics for a non-dividend-paying stock:

$$\frac{dS}{S} = r dt + \sigma dz^Q.$$

Pricing lesson for Lecture 6

In Black-Scholes-Merton, the real-world expected return μ does not enter the option price. Dynamic hedging replaces it with the risk-free rate r.

▸ Risk-neutral terminal distribution

Where We Are Going Next

Lecture 6 uses today's machinery to price options.

1. Use Ito's lemma for an option value $V(S, t)$.
2. Combine the option and stock to eliminate the dz risk.
3. The locally riskless portfolio must earn r.
4. This gives the Black-Scholes-Merton PDE.
5. Solve the PDE or use risk-neutral valuation to get option formulas.

Key bridge

The common Brownian shock in dS and dV is what makes delta hedging possible.

Checklist

After this lecture you should be able to:

- define stochastic process, Markov process, and Wiener process;
- explain square-root-time volatility scaling;
- write generalized Wiener, Ito, and GBM processes;
- simulate a GBM step and annualize volatility;
- construct correlated normal shocks;
- use the stochastic multiplication table;
- state and apply Ito's lemma;
- derive $d \ln S$ from GBM;
- state the lognormal distribution of S_T;
- explain the real-world versus risk-neutral drift change.

Practice Questions

1. If annual volatility is 24%, what is one-month volatility?
2. If z_t is Brownian motion, what is the distribution of $z_3 - z_1$?
3. For $dx = 5dt + 2dz$, find the mean and variance of $x_T - x_0$.
4. For $dS/S = 0.10dt + 0.25dz$, find the approximate one-week return distribution.
5. Use Ito's lemma to find $d(S^2)$ when $dS = \mu S dt + \sigma S dz$.
6. Derive $d \ln S$ and explain the $-\sigma^2/2$ term.
7. State the distribution of $\ln S_T$ under risk-neutral dynamics.

Section 7

Appendix Derivations

7. Appendix Derivations

Appendix: Markov Conditional Distribution

A process is Markov if the current state is a sufficient statistic for the future:

$$\mathbb{P}(X_u \in A \mid \mathcal{F}_t) = \mathbb{P}(X_u \in A \mid X_t), \qquad u > t.$$

Here $\mathcal{F}_t$ means all information generated by the process up to time t.

Implication

If two histories reach the same current value $X_t = x$, the model assigns the same future distribution to both histories.

Back to Markov slide

Appendix: Variance Scaling

Let the change over T be divided into N independent increments:

$$\Delta X_T = \sum_{i=1}^{N} \Delta X_i, \qquad \Delta X_i \sim \mathcal{N}(0, \Delta t), \qquad N\Delta t = T.$$

Then

$$\mathbb{E}[\Delta X_T] = \sum_{i=1}^{N} \mathbb{E}[\Delta X_i] = 0,$$

and independence gives

$$\mathrm{Var}(\Delta X_T) = \sum_{i=1}^{N} \mathrm{Var}(\Delta X_i) = N\Delta t = T.$$

Therefore $\mathrm{sd}(\Delta X_T) = \sqrt{T}$.

Back to variance slide

Appendix: Deterministic Growth

If there is no uncertainty in GBM,

$$dS = \mu S dt.$$

Divide by S:

$$\frac{dS}{S} = \mu dt.$$

Integrate from 0 to T:

$$\int_{S_0}^{S_T} \frac{dS}{S} = \int_0^T \mu dt.$$

Hence

$$\ln S_T - \ln S_0 = \mu T, \qquad S_T = S_0 e^{\mu T}.$$

Back to GBM slide

Appendix: Correlated Normal Construction

Draw independent $u, v \sim \mathcal{N}(0,1)$. Define

$$\varepsilon_1 = u, \qquad \varepsilon_2 = \rho u + \sqrt{1-\rho^2}\, v.$$

Means:

$$\mathbb{E}[\varepsilon_1] = 0, \qquad \mathbb{E}[\varepsilon_2] = 0.$$

Variance of ε_2:

$$\mathrm{Var}(\varepsilon_2) = \rho^2 \,\mathrm{Var}(u) + (1-\rho^2)\,\mathrm{Var}(v) = 1.$$

Covariance:

$$\mathrm{Cov}(\varepsilon_1, \varepsilon_2) = \mathrm{Cov}(u, \rho u + \sqrt{1-\rho^2}\, v) = \rho.$$

Back to correlation slide

Appendix: Why $(dz)^2 = dt$

On a small interval:

$$dz \approx \varepsilon\sqrt{dt}, \qquad \varepsilon \sim \mathcal{N}(0,1).$$

Then

$$(dz)^2 \approx \varepsilon^2 dt.$$

Since $\mathbb{E}[\varepsilon^2] = 1$, the expected size of $(dz)^2$ is dt.

Order comparison

$$(dt)^2 \text{ is order } dt^2, \qquad dt\,dz \text{ is order } dt^{3/2}, \qquad (dz)^2 \text{ is order } dt.$$

Only order-dt terms survive in the stochastic differential.

Back to rules slide

Appendix: Deriving Ito's Lemma

Start with Taylor expansion:

$$dG = G_x dx + G_t dt + \frac{1}{2} G_{xx} (dx)^2 + G_{xt} dx\, dt + \frac{1}{2} G_{tt} (dt)^2 + \cdots .$$

If

$$dx = a\, dt + b\, dz,$$

then, using the multiplication rules,

$$(dx)^2 = (a\, dt + b\, dz)^2 = b^2 (dz)^2 = b^2 dt.$$

Higher-order terms $dx\, dt$ and $(dt)^2$ vanish.
Therefore

$$dG = \left(G_x a + G_t + \frac{1}{2} G_{xx} b^2 \right) dt + G_x b\, dz.$$

Back to Ito's lemma

Appendix: Power Function Under GBM

Let $G(S) = S^n$ and

$$dS = \mu S dt + \sigma S dz.$$

The derivatives are

$$G_S = nS^{n-1}, \qquad G_{SS} = n(n-1)S^{n-2}.$$

Ito's lemma gives

$$dG = \left[n\mu S^n + \frac{1}{2} n(n-1)\sigma^2 S^n \right] dt + n\sigma S^n dz.$$

Since $G = S^n$:

$$\frac{dG}{G} = \left[n\mu + \frac{1}{2} n(n-1)\sigma^2 \right] dt + n\sigma dz.$$

Back to examples slide

Appendix: Forward Price Dynamics

Let

$$F(S,t) = Se^{r(T-t)}.$$

Derivatives:

$$F_S = e^{r(T-t)}, \qquad F_{SS} = 0, \qquad F_t = -rSe^{r(T-t)}.$$

With $dS = \mu S dt + \sigma S dz$, Ito's lemma gives:

$$dF = \left[\mu S e^{r(T-t)} - rSe^{r(T-t)}\right] dt + \sigma S e^{r(T-t)} dz.$$

Since $F = Se^{r(T-t)}$:

$$dF = (\mu - r)F dt + \sigma F dz.$$

Back to forward slide

Appendix: Deriving $d \ln S$

Let $G(S) = \ln S$. Then

$$G_S = \frac{1}{S}, \qquad G_{SS} = -\frac{1}{S^2}, \qquad G_t = 0.$$

With

$$dS = \mu S dt + \sigma S dz,$$

Ito's lemma gives:

$$d \ln S = \left[\frac{1}{S} \mu S + \frac{1}{2} \left(-\frac{1}{S^2} \right) \sigma^2 S^2 \right] dt + \frac{1}{S} \sigma S dz.$$

Therefore

$$d \ln S = \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dz.$$

Back to log-process slide

Appendix: Lognormal Terminal Distribution

From

$$d \ln S = \left(\mu - \frac{1}{2}\sigma^2\right) dt + \sigma dz,$$

integrate from 0 to T:

$$\ln S_T - \ln S_0 = \left(\mu - \frac{1}{2}\sigma^2\right) T + \sigma(z_T - z_0).$$

Since $z_T - z_0 \sim \mathcal{N}(0, T)$:

$$\ln S_T \sim \mathcal{N}\left(\ln S_0 + \left(\mu - \frac{1}{2}\sigma^2\right) T, \sigma^2 T\right).$$

Equivalent simulation formula:

$$S_T = S_0 \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right) T + \sigma\sqrt{T}\varepsilon\right].$$

Back to terminal distribution

Back to simulation

Appendix: Risk-Neutral Terminal Distribution

Under the risk-neutral measure for a non-dividend-paying stock:

$$\frac{dS}{S} = r\,dt + \sigma\,dz^Q.$$

Repeat the log transformation:

$$d\ln S = \left(r - \frac{1}{2}\sigma^2\right)dt + \sigma\,dz^Q.$$

Therefore:

$$\ln S_T \sim \mathcal{N}\left(\ln S_0 + \left(r - \frac{1}{2}\sigma^2\right)T, \sigma^2 T\right).$$

This is the terminal distribution used in risk-neutral valuation for the Black-Scholes-Merton formula.

Back to risk-neutral slide